

SQL Review Research Activity

Part 1: SQL Fundamentals Review

1. What is SQL?

SQL (Structured Query Language) is a standard programming language for accessing and manipulating databases.

2. Why is SQL important in database systems?

SQL is important because it:

- Handles data management and retrieval – almost all database systems use sql to add, remove, retrieve, and update information
- Handle massive datasets: able to queries millions row of data
- Gives data consistency, integrity and security: allows for definition of constraints that ensure data accuracy and provide control over who can view or modify data
- Help increase efficiency: For non-technical roles like marketing or finance, knowing SQL allows employees to pull their own data, saving time and removing reliance on busy IT teams
- Data Organization: allows for data to be in a structure format avoiding errors and making it easier to know about the data of the system itself

3. Explain the difference between:

- DDL: (Data Definition Language) consists of SQL commands that can be used for defining, altering and deleting database structures such as tables, indexes and schemas. It simply deals with descriptions of the database schema and is used to create and modify the structure of database objects in the database.

Example:

```
CREATE TABLE employees (  
  employee_id INT PRIMARY KEY,  
  first_name VARCHAR(50),  
  last_name VARCHAR(50),  
  hire_date DATE  
);
```

- DML: (Data Manipulation Language) are commands used to manipulate the data stored in database tables. With DML, you can insert new records, update existing ones, delete unwanted data or retrieve information.

Example:

```
INSERT INTO employees (first_name, last_name, department)  
VALUES ('Jane', 'Smith', 'HR');
```

- DCL: (Data Control Language) includes commands such as GRANT and REVOKE which mainly deal with the rights, permissions and other controls of the database system. These commands are used to control access to data in the database by granting or revoking permissions.

Example:

```
GRANT SELECT, UPDATE ON employees TO user_name;
```

- TCL: (Transaction Control Language) group a set of tasks into a single execution unit. Each transaction begins with a specific task and ends when all the tasks in the group are successfully completed. If any of the tasks fail, transaction fails. Therefore, a transaction has only two results: success or failure.

Example:

BEGIN TRANSACTION;

UPDATE employees SET department = 'Marketing' WHERE department = 'Sales';

SAVEPOINT before_update;

UPDATE employees SET department = 'IT' WHERE department = 'HR';

ROLLBACK TO SAVEPOINT before_update;

COMMIT;

4. Describe the purpose of the following SQL clauses/commands:
 - SELECT: It is used to retrieve data from the database
 - WHERE: Filters rows before any grouping or aggregation
 - ORDER BY: Sorts the result set by one or more columns in ascending or descending order
 - GROUP BY: Groups rows that have the same values in specified columns
 - HAVING: Filters the results of GROUP BY (The HAVING clause was added to SQL because the WHERE keyword cannot be used with aggregate functions.)
 - JOIN: is used to combine rows from two or more tables, based on a related column between them.

Part 2: Query Analysis

Query A

`SELECT * FROM employees;`

What it does: Retrieves all columns and all records/information from the employees table

Real world use case: Usually used by HR to view the complete list of employees for reporting or auditing process as all info and records on the employees are needed

Query B

`SELECT department, AVG(salary)
FROM employees
GROUP BY department;`

What it does: Groups employees by department and calculate the average salary from each group

Real world use case: Used by management to analyze salary distribution across department to ensure fairness.

Query C

```
SELECT c.name, o.total  
FROM customers c  
JOIN orders o ON c.customer_id = o.customer_id;
```

What it does: Join the customers table and orders tables to display costumers' name with their order total

Real world use case: Used in sales systems to track costumer purchases and analyze behavior

Part 3: SQL in Real-World Applications

Choose **three (3) industries** and explain how SQL is used in each.

Possible industries include:

- Banking
- Healthcare
- E-commerce
- Education
- Government
- Logistics
- Etc.

Banking:

1. Types of data stored: Customer info, account balance, transaction history and details, loan details.
2. How SQL is used to manage/analyze the data: Used to retrieve account details, process transactions, detect scam/fraud patterns.
3. Example SQL-related tasks in that industry: Checking account balance, recording transaction or deposits and withdrawals, generate monthly expenses and income statement.

Healthcare:

1. Types of data stored: Patient records, medical history, billing info, appointments
2. How SQL is used to manage/analyze the data: Manage patient data, track treatments, generate medical analysis reports
3. Example SQL-related tasks in that industry: Retrieve patient history, schedule appointment, analyze disease trend (what is most common at the moment)

Logistic:

1. Types of data stored: Shipment details, delivery status, customer info, supplier info, inventory and warehouse data, transportation routes and vehicle details
2. How SQL is used to manage/analyze the data: Manage orders, track location of goods, manage stocks, analyze operational cost
3. Example SQL-related tasks in that industry: Tracking current status of shipment, updating delivery process, manage warehouse inventory levels, generate reports on delivery performances

Part 4: Reflection

Answer the following:

1. Which SQL concept do you find most challenging? Why?
The join operations especially with more than 2 tables bcus it is difficult to understand how different tables can be related to one another and how or where it is connected, combining them without errors is a hard part.
2. Which SQL concept do you believe is most important for industry use? Why?
In my opinion, it is the select and where bcus it is the most used concept to retrieve data quickly and efficiently. That is the core of every database operation system.
3. How does SQL support data-driven decision making?
SQL allows a large amount of data to be analyzed quickly and extract insights or exact information needed. By using queries, we can identify trends, patterns of sales, measure performance, make decision based on real time data/report that is accurate and consistent rather than assumptions.

Name: Chey Naryvety

Group: 4

ID: IDTB110199